

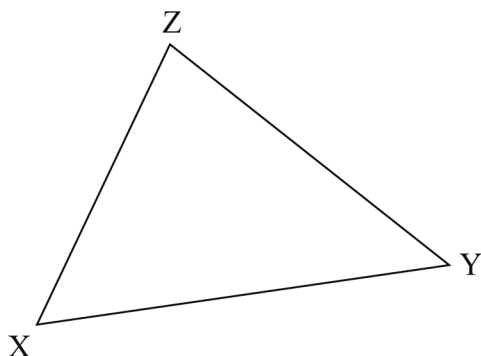


3.3.2 Mathematics Alt. A Paper 2 (121/2)

SECTION I (50 marks)

Answer **all** the questions in this section in the spaces provided.

- 1 The sum of n terms of the sequence; 3, 9, 15, 21, ... is 7500. Determine the value of n . (3 marks)
- 2 A quadratic curve passes through the points $(-2, 0)$ and $(1, 0)$. Find the equation of the curve in the form $y = ax^2 + bx + c$, where a , b and c are constants. (2 marks)
- 3 Make d the subject of the formula,
- $$P = \frac{1}{2}mn^2 - \frac{qd^2}{n}$$
- (3 marks)
- 4 Solve the equation
- $$2 \log x - \log (x - 2) = 2 \log 3.$$
- (3 marks)
- 5 (a) Using a pair of compasses and ruler only, construct an escribed circle to touch side XZ of triangle XYZ drawn below. (3 marks)



- (b) Measure the radius of the circle. (1 mark)
- 6 The equation of a circle is given by $x^2 + 4x + y^2 - 2y - 4 = 0$. Determine the centre and radius of the circle. (3 marks)
- 7 (a) Expand $(1 - x)^5$. (1 mark)
- (b) Use the expansion in (a) up to the term in x^3 to approximate the value of $(0.98)^5$. (2 marks)
- 8 The position vectors of points F, G and H are $\mathbf{f}$, $\mathbf{g}$ and $\mathbf{h}$ respectively. Point H divides FG in the ratio 4:–1. Express $\mathbf{h}$ in terms of $\mathbf{f}$ and $\mathbf{g}$. (2 marks)



- 9 Two machines, M and N produce 60% and 40% respectively of the total number of items manufactured in a factory. It is observed that 5% of the items produced by machine M are defective while 3% of the items produced by machine N are defective. If an item is selected at random from the factory, find the probability that it is defective. (3 marks)
- 10 Two taps A and B can each fill an empty tank in 3 hours and 2 hours respectively. A drainage tap R can empty the full tank in 6 hours. Taps A and R are opened for 5 hours then closed.
- (a) Determine the fraction of the tank that is still empty. (2 marks)
- (b) Find how long it would take to fill the remaining fraction of the tank if all the three taps are opened. (2 marks)
- 11 Simplify the expression $\frac{\sqrt{48}}{\sqrt{5} + \sqrt{3}}$, leaving the answer in the form $a\sqrt{b} + c$ where a, b and c are integers. (3 marks)
- 12 A point P moves inside a sector of a circle, centre O, and chord AB such that $2\text{ cm} < OP \leq 3\text{ cm}$ and angle $APB = 65^\circ$. Draw the locus of P. (4 marks)
- 13 The table below shows income tax rates in a certain year.

Monthly income in Kenya shillings	Tax rate in each shilling
Up to 9 680	10%
from 9 681 to 18 800	15%
from 18 801 to 27 920	20%
from 27 921 to 37 040	25%
over 37 040	30%

In that year, a monthly personal tax relief of Ksh 1 056 was allowed. Calculate the monthly income tax paid by an employee who earned a monthly salary of Ksh 32 500. (4 marks)

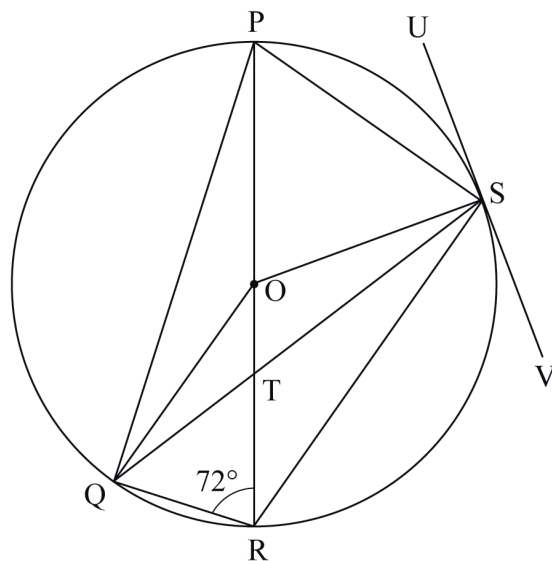
- 14 Solve the equation $6 \cos^2 x + 7 \sin x - 8 = 0$ for $0^\circ \leq x \leq 90^\circ$. (4 marks)
- 15 The positions of two towns are $(2^\circ\text{S}, 30^\circ\text{E})$ and $(2^\circ\text{S}, 37.4^\circ\text{E})$. Calculate, to the nearest km, the shortest distance between the two towns. (Take the radius of the earth to be 6 370 km) (2 marks)
- 16 The vertices of a triangle T are A(1, 2), B(4, 2) and C(3, 4). The vertices of triangle T', the image of T are $A'(\frac{1}{2}, 1)$, $B'(2, 1)$ and $C'(\frac{3}{2}, 2)$.
- Determine the transformation matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ that maps T onto T'. (3 marks)



SECTION II (50 marks)

Answer only **five** questions from this section in the spaces provided.

- 17 The Hire Purchase (H.P.) price of a public address system was Ksh 276 000. A deposit of Ksh 60 000 was paid followed by 18 equal monthly instalments. The cash price of the public address system was 10% less than the H.P. price.
- (a) Calculate:
- (i) the monthly instalment; (2 marks)
 - (ii) the cash price. (2 marks)
- (b) A customer decided to buy the system in cash and was allowed a 5% discount on the cash price. He took a bank loan to buy the system in cash. The bank charged compound interest on the loan at the rate of 20% p.a. The loan was repaid in 2 years. Calculate the amount repaid to the bank by the end of the second year. (3 marks)
- (c) Express as a percentage of the Hire Purchase price, the difference between the amount repaid to the bank and the Hire Purchase price. (3 marks)
- 18 In the figure below, PR is a diameter of the circle centre O. Points P, Q, R and S are on the circumference of the circle. Angle $PRQ = 72^\circ$, $QS = QP$ and line USV is a tangent to the circle at S.



Giving reasons, calculate the size of:

- (a) $\angle QPR$; (2 marks)
- (b) $\angle PQS$; (2 marks)
- (c) $\angle OQS$; (2 marks)
- (d) $\angle RTS$; (2 marks)
- (e) $\angle RSV$. (2 marks)

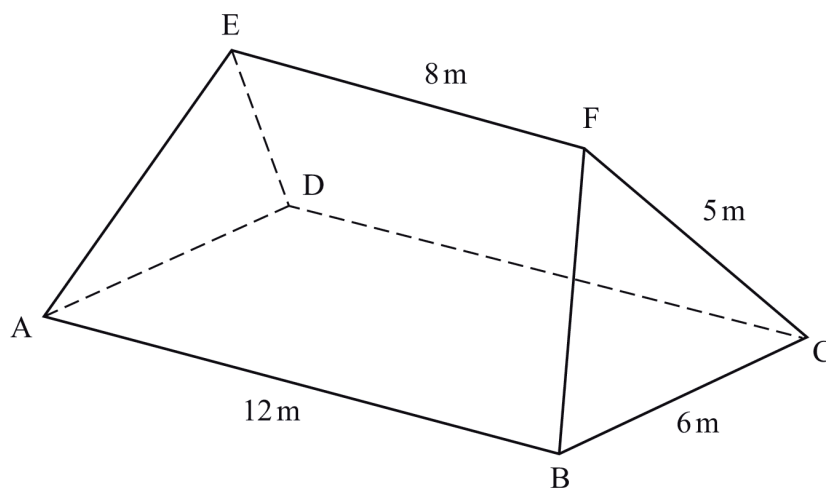


- 19 (a) Complete the table below for $y = x^3 + 4x^2 - 5x - 5$. (2 marks)

x	- 5	- 4	- 3	- 2	- 1	0	1	2
$y = x^3 + 4x^2 - 5x - 5$			19			- 5		

- (b) On the grid provided, draw the graph of $y = x^3 + 4x^2 - 5x - 5$ for $- 5 \leq x \leq 2$. (3 marks)
- (c) (i) Use the graph to solve the equation $x^3 + 4x^2 - 5x - 5 = 0$. (2 marks)
- (ii) By drawing a suitable straight line on the graph, solve the equation $x^3 + 4x^2 - 5x - 5 = - 4x - 1$. (3 marks)

- 20 The figure ABCDEF below represents a roof of a house. $AB = DC = 12$ m, $BC = AD = 6$ m, $AE = BF = CF = DE = 5$ m and $EF = 8$ m.



- (a) Calculate, correct to 2 decimal places, the perpendicular distance of EF from the plane ABCD. (3 marks)
- (b) Calculate the angle between:
- (i) the planes ADE and ABCD; (2 marks)
- (ii) the line AE and the plane ABCD, correct to 1 decimal place; (2 marks)
- (iii) the planes ABFE and DCFE, correct to 1 decimal place. (3 marks)



- 21** (a) Complete the table below, giving the values correct to 1 decimal place. (2 marks)

x°	0	40	80	120	160	200	240
$2 \sin (x + 20)^\circ$	0.7		2.0		0.0		- 2.0
$\sqrt{3} \cos x$	1.7	1.3		- 0.9		- 1.6	

- (b) On the grid provided, using the same scale and axes, draw the graphs of $y = 2 \sin (x + 20)^\circ$ and $y = \sqrt{3} \cos x$ for $0^\circ \leq x \leq 240^\circ$. (5 marks)
- (c) Use the graphs drawn in (b) above to determine:
- (i) the values of x for which $2 \sin (x + 20) = \sqrt{3} \cos x$; (2 marks)
- (ii) the difference in the amplitudes of $y = 2 \sin (x + 20)$ and $y = \sqrt{3} \cos x$. (1 mark)

- 22** Three quantities R, S and T are such that R varies directly as S and inversely as the square of T.

- (a) Given that $R = 480$ when $S = 150$ and $T = 5$, write an equation connecting R, S and T. (4 marks)
- (b) (i) Find the value of R when $S = 360$ and $T = 1.5$. (2 marks)
- (ii) Find the percentage change in R if S increases by 5% and T decreases by 20%. (4 marks)

- 23** The equation of a curve is given by $y = 5x - \frac{1}{2}x^2$.

- (a) On the grid provided, draw the curve of $y = 5x - \frac{1}{2}x^2$ for $0 \leq x \leq 6$. (3 marks)
- (b) By integration, find the area bounded by the curve, the line $x = 6$ and the x-axis. (3 marks)
- (c) (i) On the same grid as in (a), draw the line $y = 2x$. (1 mark)
- (ii) Determine the area bounded by the curve and the line $y = 2x$. (3 marks)



- 24 The table below shows marks scored by 42 students in a test.

35	49	69	57	58	75	48
40	46	86	47	81	67	63
56	80	36	62	49	46	26
41	58	68	73	65	59	72
64	70	64	54	74	33	51
73	25	41	61	56	57	28

- (a) Starting with the mark of 25 and using equal class intervals of 10, make a frequency distribution table. (2 marks)
- (b) On the grid provided, draw the ogive for the data. (4 marks)
- (c) Using the graph in (b) above, estimate:
- (i) the median mark; (2 marks)
 - (ii) the upper quartile mark. (2 marks)